1. Clinical Question & Core Hypothesis

Clinical Question

Tier 1 synthesis highlights a paradox: MSH6/MMR-driven MSI should increase neoantigen load and immunogenicity, yet MSI/MSH6 status alone does not universally predict favorable biomarker-guided outcomes or immunotherapy benefit in MIBC. Why does high mutational neoantigen potential sometimes fail to translate into effective anti-tumor immunity? 2) Tier 1 paths used: I built on (a) the MSH6-as-prognostic-factor path because MSH6 loss is the canonical MMR perturbation linking genomic instability to neoantigen generation, and (b) the MSI→AI-biomarkers and (c) immunotherapy-biomarker paths because they frame MSI as an immunogenic signal whose clinical utility depends on immune-context features that imaging/AI can capture. These together motivate testing an immune-organization requirement for MSI-driven benefit. 3) Tier 2 support: PTGDS, pZVR, PSA and HDL-C provide orthogonal, cross-cancer biomarker signals that could modulate tumor inflammation, stromal composition or systemic milieu and thus act as secondary modifiers of the MSI–immune axis; they justify exploring confounders and effect modifiers beyond MMR. 4) Endpoint choice: I selected therapy_response (specifically immunotherapy/ICI response proxy) because the mechanistic question centers on why neoantigen-generating MMR defects do not always yield clinical benefit from immune-based therapies—therapy_response is the most biologically direct and actionable endpoint for this question. 5) Deliberate exclusions: I excluded Tier 3 concepts (Immunoscore, HDL-C as primary mechanistic anchors) from the core claim to keep the hypothesis sharply focused on the interaction between tumor genomic immunogenicity (MSH6/MSI) and local immune organization (TLS); Tier 2 candidates were retained as secondary_modifiers for downstream analyses?

Working Hypothesis

MSH6-deficient/MSI-high muscle-invasive bladder tumors realize their neoantigen-driven immunogenic potential to elicit a clinically meaningful immunotherapy response only when the tumor microenvironment contains TLS-like lymphoid aggregates. Specifically, MSI-H/MSH6 loss will be associated with higher therapy_response to PD-1/PD-L1 inhibitors primarily in TLS-rich tumors, whereas MSI-H/MSH6 loss in TLS-poor tumors will show attenuated or no improved response despite elevated neoantigen load. Tier2 modifiers (PTGDS, pZVR, PSA, HDL-C) modulate this axis by shaping inflammatory signaling, stromal remodeling, or systemic metabolic milieu, thereby influencing TLS formation/maintenance and local antigen presentation. Consequently, the predictive value of MSI/MSH6 for immunotherapy response will be strongest in TLS-high tumors and will be further stratified by Tier2 biomarker levels, with a measurable interaction between MSI/MSH6 status, TLS presence, and Tier2 modifiers on therapy_response.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	MSH6, MSI
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6-as-prognostic-factor path because MSH6 loss is the canonical MMR perturbation linking genomic instability to neoantigen generation, and (b) the MSI→AI-biomarkers and (c) immunotherapy-biomarker paths because they frame MSI as an immunogenic signal whose clinical utility depends on immune-context features that imaging/AI can capture.

MSI: MSI→AI-biomarkers and (c) immunotherapy-biomarker paths because they frame MSI as an immunogenic signal whose clinical utility depends on immune-context features that imaging/AI can capture.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PTGDS	Supporting modifier from knowledge graph
Secondary	pZVR	Supporting modifier from knowledge graph
Secondary	PSA	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PTGDS: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- pZVR: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.